

A BAP Obstruction for Filter Bases over UBP Filters

Run `fa_banach_001`, target arXiv:2509.20002

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Source And Target

The source paper is V. Kadets and M. Manskova, *Norms of partial sums operators for a basis with respect to a filter*, arXiv:2509.20002. Near the end of the paper the authors write that one of the most intriguing open problems about $\mathcal{F}$ -bases is whether every separable Banach space has an $\mathcal{F}$ -basis for some free filter $\mathcal{F}$, and, for a fixed filter $\mathcal{F}$, whether a given space has an $\mathcal{F}$ -basis. They mention the statistical convergence filter as an important concrete case still outside their slow-filter negative theorem.

This packet records a partial obstruction for a different large class of filters: Banach-UBP filters in the sense of De Bondt–Vernaeve.

Definitions

Let $\mathcal{F}$ be a free filter on $\mathbb{N}$. A sequence (e_n) in a Banach space X is an $\mathcal{F}$ -basis if for every $x \in X$ there is a unique scalar sequence (a_n) such that

$$\mathcal{F}\text{-}\lim_n \sum_{k=1}^n a_k e_k = x.$$

When the coordinate functionals $e_n^*(x) = a_n$ are continuous, the partial sums

$$S_n x = \sum_{k=1}^n e_k^*(x) e_k$$

are continuous finite-rank operators and $S_n x \rightarrow_{\mathcal{F}} x$ for every $x \in X$.

Following De Bondt–Vernaeve, $\mathcal{F}$ is a Banach-UBP filter if the uniform $\mathcal{F}$ -boundedness principle holds for every Banach domain. In the normed operator setting this means: every pointwise $\mathcal{F}$ -bounded sequence $(T_n) \subset L(X, Y)$ is $\mathcal{F}$ -bounded in operator norm.

Idea Of The Proof

The partial sums of a continuous-coordinate $\mathcal{F}$ -basis are finite-rank approximants to the identity along $\mathcal{F}$. Pointwise $\mathcal{F}$ -convergence makes them pointwise $\mathcal{F}$ -bounded. The Banach-UBP property upgrades this to a uniform operator-norm bound on an $\mathcal{F}$ -large set of indices. Intersect that large set with the finitely many approximation sets needed for a compact net; one of the remaining finite-rank partial sums approximates the whole compact set with the same uniform bound. That is the bounded approximation property.

Theorem 1. *Let $\mathcal{F}$ be a Banach-UBP filter. If a Banach space X has an $\mathcal{F}$ -basis whose coordinate functionals are continuous, then X has the bounded approximation property.*

Proof. Let (e_n) be the $\mathcal{F}$ -basis and let S_n be its partial sum operators. For each $x \in X$, $S_n x \rightarrow_{\mathcal{F}} x$, hence $(S_n x)$ is $\mathcal{F}$ -bounded in X . Thus (S_n) is pointwise $\mathcal{F}$ -bounded in $L(X, X)$.

Since $\mathcal{F}$ is a Banach-UBP filter, (S_n) is $\mathcal{F}$ -bounded in operator norm. Therefore there are a set $A \in \mathcal{F}$ and a constant $C < \infty$ such that

$$\sup_{n \in A} \|S_n\| \leq C.$$

Let $K \subset X$ be compact and fix $\varepsilon > 0$. Choose $x_1, \dots, x_m \in K$ such that every point of K is within

$$\delta = \frac{\varepsilon}{3(C+1)}$$

of some x_j . For each j , the set

$$G_j = \{n \in \mathbb{N} : \|S_n x_j - x_j\| < \varepsilon/3\}$$

belongs to $\mathcal{F}$. Hence

$$A \cap G_1 \cap \dots \cap G_m \in \mathcal{F}$$

and is nonempty. Pick n in this intersection. Then S_n is finite rank, $\|S_n\| \leq C$, and for any $x \in K$, choosing x_j with $\|x - x_j\| < \delta$, we have

$$\begin{aligned} \|S_n x - x\| &\leq \|S_n(x - x_j)\| + \|S_n x_j - x_j\| + \|x_j - x\| \\ &\leq C\delta + \varepsilon/3 + \delta < \varepsilon. \end{aligned}$$

Thus finite-rank operators with a common norm bound approximate the identity uniformly on arbitrary compact subsets of X . This is the bounded approximation property. $\square$

Corollary 1. *Let X be a separable Banach space without the approximation property. Then X has no $\mathcal{F}$ -basis with continuous coordinate functionals for any Banach-UBP filter $\mathcal{F}$.*

Proof. The bounded approximation property implies the approximation property. Therefore the theorem rules out such an $\mathcal{F}$ -basis on any space failing AP, for example Enflo's separable counterexample to the approximation problem. $\square$

Corollary 2. *If $\mathcal{F}$ is both analytic and a Banach-UBP filter, then every separable Banach space without the approximation property has no $\mathcal{F}$ -basis.*

Proof. De Rancourt–Kania–Swaczyna prove that for analytic filters the coordinate functionals of every $\mathcal{F}$ -basis are continuous. The preceding corollary then applies. $\square$

Limitations

This is a partial obstruction, not a full solution of the source problem. It does not decide whether every separable Banach space has an $\mathcal{F}$ -basis for some free filter. It also does not decide the fixed-filter problem for the statistical convergence filter. De Bondt–Vernaëve record that the asymptotic density filter is weak P^+ but neither P^+ nor rapid $^+$, so it is outside the Banach-UBP class even though it is analytic and hence has automatic coordinate continuity by de Rancourt–Kania–Swaczyna.

The separate endpoint problem in the source's description of $\mathcal{A}(w^*, \ell_p^*, \mathcal{F})$ for $1 < p < 2$ also remains open after this pass.

Novelty Check

The run indexes were searched for arXiv:2509.20002, the paper title, **F-basis**, **filter basis**, **bounded approximation property**, **BAP**, **UBP**, and **approximation property**. No prior packet or attempt for this target was found. Web searches on 2026-07-01 for exact and close phrases such as “filter basis bounded approximation property”, “F-basis bounded approximation property”, and “F-basis Uniform Boundedness Principle filter Banach” did not reveal a direct prior statement of this obstruction.

References

- [1] V. Kadets and M. Manskova, *Norms of partial sums operators for a basis with respect to a filter*, arXiv:2509.20002.
- [2] M. De Bondt and H. Vernaëve, *Filter-dependent versions of the Uniform Boundedness Principle*, J. Math. Anal. Appl. 504 (2021), arXiv:2001.01663.
- [3] N. de Rancourt, T. Kania, and J. Swaczyna, *Continuity of coordinate functionals of filter bases in Banach spaces*, J. Funct. Anal. 284 (2023), Article 109869, arXiv:2203.15123.
- [4] P. Enflo, *A counterexample to the approximation problem in Banach spaces*, Acta Math. 130 (1973), 309–317.